Total No. of Questions : 8]

SEAT No. :

PE-2744

[Total No. of Pages : 2

[6583]-325

T.E. (Mechanical / Automobile)

Honors in E-Vehicle Technology

(2019 Pattern) (Semester - V) (302031MJ)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates :

1) ***Solve Q.1 or Q. 2, Q.3 or Q.4, Q.5 or Q.6 and Q.7or Q.8.***
2) ***Figures to the right indicate full marks***
3) ***Draw the neat sketch wherever necessary.***

Q1) a) Explain working of Li-ion battery cell. Also mention two applications. **[9]**

b) What are the different chemistries of li-ion batteries? Mention in short significance of each of them for electric vehicle. **[8]**

OR

Q2) a) Explain the following parameters of li-ion batteries **[9]**

i) State of Charge

ii) Energy density

iii) C Rating

b) What is the type of charging li-ion battery? Explain the precautions needs to be taken while charging. **[8]**

Q3) a) Explain Nickel-Metal Hydride Batteries with advantages, disadvantages and applications **[9]**

b) Write a short not on need of thermal management system in e-vehicles? Explain with mechanism **[9]**

OR

P.T.O.

Q4) a) Explain Li-ion super capacitor with advantages, disadvantages and applications **[9]**

b) Explain lead acid batteries with advantages, disadvantages and applications **[9]**

Q5) a) Write a short note on Mechanical and electrical connections of motors. **[8]**

b) Explain following types of motor. **[9]**

i) Series Motor

ii) Parallel/Shunt DC Motor

iii) Compound DC motor

OR

Q6) a) Write a short note on Output Characteristics of Motor Drives in EVs? **[8]**

b) Explain with neat sketch working of BLDC Motor. Give two application **[9]**

Q7) a) Write a short note on Battery Standards in electric vehicles with any two types? **[10]**

b) Write a short note on intelligent transport system with its advantages? **[8]**

OR

Q8) a) What is battery swapping? Explain with Advantages And Challenges Of Battery Swapping p **[8]**

b) Explain the significance of implementation of IOT in electric vehicle on basis of wireless sensor network with neat sketch wherever required. **[10]**